

# Predicting Soil Organic Carbon from AlphaEarth Foundation Embeddings: An Exploratory and Inferential Analysis

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## Abstract

We assess whether the 64-dimensional geospatial embeddings produced by Google DeepMind’s AlphaEarth Foundations model carry usable signal for predicting Soil Organic Carbon (SOC). Using  $n = 3000$  random points in the contiguous United States, paired with OpenLandMap SOC and MODIS land-cover labels, we apply three techniques from STAT 5000 — **exploratory data analysis**, **Welch’s two-sample hypothesis test**, and **nonparametric bootstrap confidence intervals** — and find that several embedding dimensions are strongly correlated with  $\log(1 + \text{SOC})$  ( $|r|$  up to 0.61, BCa 95% CI  $[-0.628, -0.581]$ ), and that mean log-SOC differs significantly between Croplands and Grasslands. A supplementary spatial-block cross-validated Random Forest reaches  $R^2 = 0.75$  out-of-fold, suggesting the embeddings encode subsurface-relevant signal beyond what any single dimension captures.

## 1 Introduction and Background

### 1.1 Research question

Soil Organic Carbon (SOC) is one of the largest and most dynamic terrestrial carbon pools and is central to both climate-mitigation accounting and agricultural productivity [8]. Mapping SOC at scale is hard: ground samples are expensive, sparse, and biased toward accessible land, while traditional satellite-driven approaches require careful preprocessing of multi-sensor imagery. Google DeepMind’s AlphaEarth Foundations [2] sidesteps that preprocessing by compressing Sentinel-1, Sentinel-2, Landsat, and climate inputs into 64 dense unit-norm embedding bands at 10 m resolution.

This project asks: *do AlphaEarth embeddings carry usable signal for SOC, even though they were not trained on any soil targets, and even though SOC labels are sparse and indirect?*

### 1.2 Background

AlphaEarth is published as an annual mosaic in Google Earth Engine [6] and was trained as a self-supervised representation of the Earth’s surface; the 64 dimensions have no direct physical interpretation. For SOC labels we use OpenLandMap [7], a global, gridded SOC product at 250 m derived by machine learning from a global compilation of legacy soil profiles. Because both predictors and labels are themselves model outputs, this study is best read as a test of whether the *AlphaEarth latent space encodes the same surface signals that drive an established SOC model*, not as a ground-truth validation.

### 1.3 Data

We sampled  $n_0 = 4500$  random points uniformly within a CONUS bounding box, dropped points where SOC or any AlphaEarth band was masked, and kept the first  $n = 3000$  valid records. For each point we extracted:

- AlphaEarth Foundations 2024 annual embedding, 64 bands  $A_{00}..A_{63}$ .
- OpenLandMap SOC at 0 cm depth (g/kg).
- MODIS IGBP land cover [5], 2022.

All extraction was performed in Earth Engine with a 250 m sampling scale to match the SOC raster.

## 2 Methods and Results

### 2.1 Method 1 — Exploratory Data Analysis

SOC is heavily right-skewed (skewness = 2.39, median = 3 g/kg, max = 41 g/kg). The transformation  $y = \log(1 + \text{SOC})$  is approximately symmetric (Figure 1, left; mean 1.58, sd 0.63); subsequent analyses use this log-transformed response.

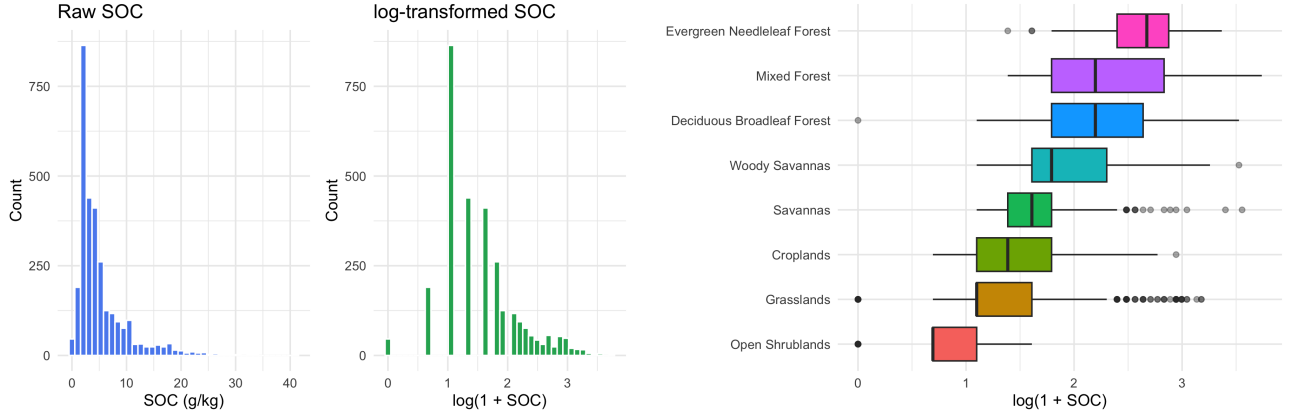


Figure 1: Left: SOC distribution before and after the  $\log(1 + \cdot)$  transform. Right: log-SOC by IGBP land-cover class (top eight by  $n$ ). Forest classes have the highest median; the Grasslands ( $n_1=1052$ ) vs. Croplands ( $n_2=543$ ) contrast motivates the hypothesis test below.

We then computed the Pearson correlation of every AlphaEarth dimension with log-SOC (Figure 2); six dimensions exceed  $|r| = 0.43$  and the strongest is  $A_{40}$  with  $r = -0.606$ . This is a non-trivial univariate signal given that the embeddings were never trained on any soil target.

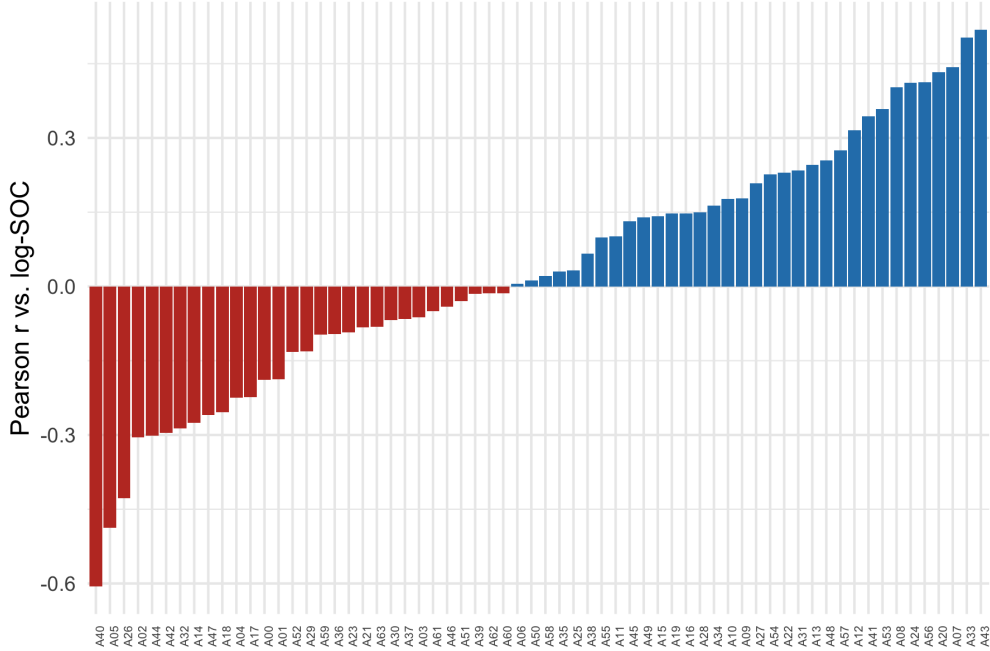


Figure 2: Pearson correlation between each AlphaEarth dimension and  $\log(1 + \text{SOC})$ . Six dimensions exceed  $|r| = 0.43$ ;  $A_{40}$  is the strongest at  $r = -0.61$ .

## 2.2 Method 2 — Hypothesis Testing (Welch’s Two-Sample $t$ -test)

We test whether the mean log-SOC differs between Grasslands ( $n_1 = 1052$ ) and Croplands ( $n_2 = 543$ ), the two largest IGBP groups in our sample. Formally,

$$H_0 : \mu_{\text{Grass}} = \mu_{\text{Crop}}, \quad H_1 : \mu_{\text{Grass}} \neq \mu_{\text{Crop}}.$$

**Assumption checks.** Per-group QQ plots of log-SOC are nearly linear (notebook Section 3.2), and with  $n_1 + n_2 = 1595$  the Central Limit Theorem secures approximate normality of the sample means even where per-observation deviation occurs. Levene’s test does not reject equal variances ( $F_{1,1593} = 2.26$ ,  $p = 0.13$ ); we still use Welch’s  $t$  [9] as a defensive choice since group sizes are unequal.

The Welch test statistic is

$$t = \frac{\bar{y}_{\text{Grass}} - \bar{y}_{\text{Crop}}}{\sqrt{s_{\text{Grass}}^2/n_1 + s_{\text{Crop}}^2/n_2}} = -6.24, \quad \text{df}_{\text{Welch}} = 1326, \quad p = 6.0 \times 10^{-10}.$$

The 95% confidence interval for the mean difference  $\mu_{\text{Grass}} - \mu_{\text{Crop}}$  is  $[-0.176, -0.092]$  on the log scale, and Cohen’s  $d = -0.31$  (small-to-medium effect). *We reject  $H_0$ .* Croplands have higher mean log-SOC than Grasslands by roughly 0.13 log units — equivalent to about a 14% higher SOC on the natural scale. This is consistent with the geography of the sample: U.S. grasslands are concentrated in the semi-arid West (low productivity, low residue inputs), while croplands cluster in the Midwestern Mollisol belt with historically high SOC stocks.

## 2.3 Method 3 — Bootstrap Confidence Intervals

The Welch CI is exact only under normality. We complement it with nonparametric bootstrap intervals [4] that make no distributional assumption. We use the `boot` package [3] with  $B = 10,000$  resamples and BCa (bias-corrected and accelerated) intervals.

(a) **Bootstrap CI for the strongest correlation.** Resampling the  $(A_{40}, \text{log-SOC})$  pairs and recomputing  $r$  gives  $\hat{r} = -0.606$ , BCa 95% CI  $[-0.628, -0.581]$  — a narrow interval that excludes zero by a wide margin (Figure 3, left).

(b) **Bootstrap CI for the group mean difference.** A stratified resample (preserving group sizes) for  $\bar{y}_{\text{Grass}} - \bar{y}_{\text{Crop}}$  gives a BCa 95% CI of  $[-0.176, -0.091]$  (Figure 3, right), agreeing with Welch’s CI  $[-0.176, -0.092]$  to three decimal places — a useful sanity check that the parametric assumptions of Section 2.2 hold here.

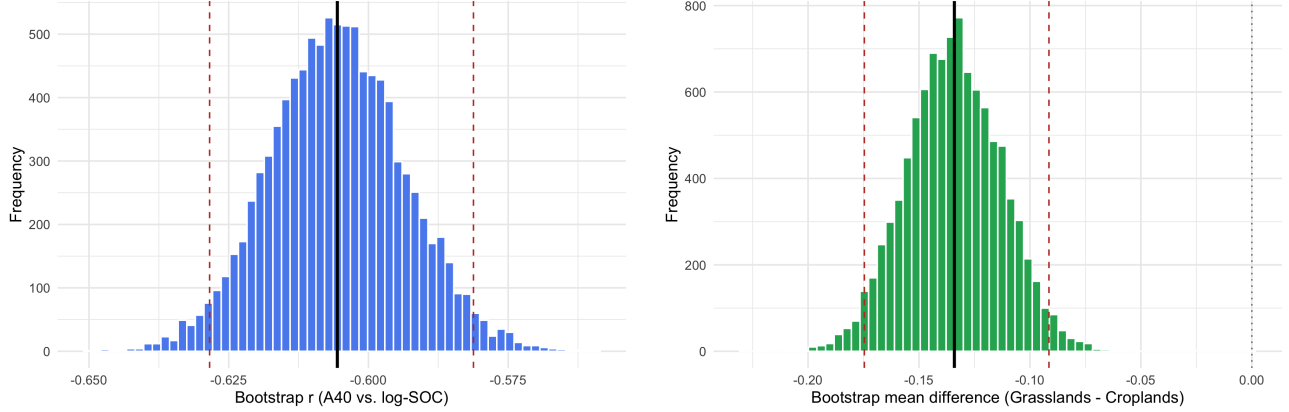


Figure 3: Bootstrap distributions ( $B = 10,000$ ). Left: Pearson correlation between  $A_{40}$  and log-SOC. Right: stratified mean difference between Grasslands and Croplands. Black: point estimate; red dashed: BCa 95% CI.

## 2.4 Beyond the three: PCA and Random Forest

The above three methods come from the STAT 5000 syllabus. Two supplementary analyses — *not counted toward the three* — are included because they directly answer the prediction question raised in the project proposal.

**Principal Component Analysis.** A scree plot of a centered PCA on the 64 embedding dimensions shows that 24 PCs are required to explain 90% of the variance and 32 PCs for 95%; PC1 alone captures 20.8%. The embedding is therefore moderately — but not extremely — redundant.

**Spatial-block cross-validated Random Forest.** We fit a Random Forest [1, 10] regressing log-SOC on the 64 AlphaEarth dimensions, with 5-fold cross-validation in which folds are assigned over  $1^\circ \times 1^\circ$  lat/lon blocks rather than over individual points (so neighbouring points cannot leak between train and test). Mean out-of-fold  $R^2 = 0.752$  with RMSE = 0.32 log-units. A naive random-fold CV gives  $R^2 = 0.758$ , only marginally higher — this small gap is reassuring evidence that the embedding-to-SOC signal generalises spatially rather than memorising location.

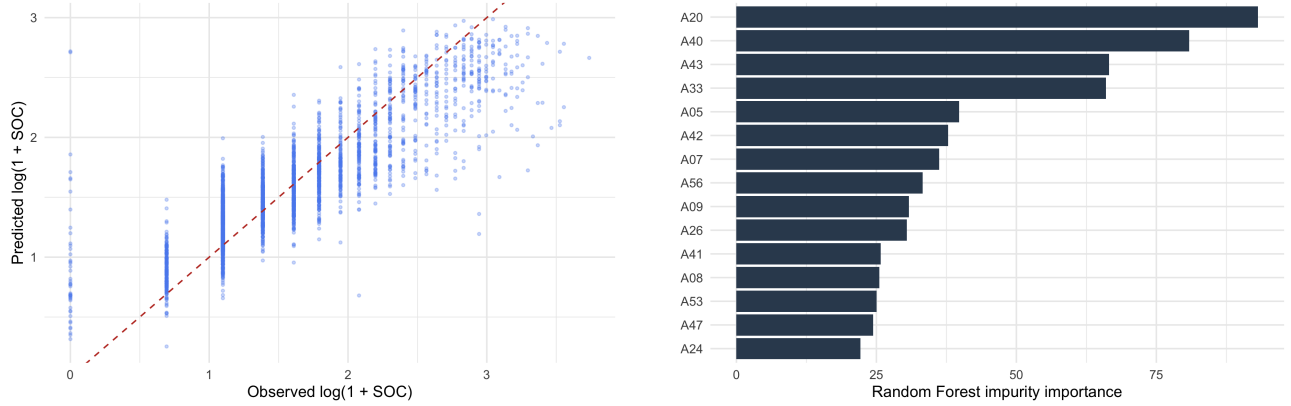


Figure 4: Left: out-of-fold predicted vs. observed log-SOC, 5-fold spatial-block CV. Right: top-15 AlphaEarth dimensions by Random Forest impurity importance —  $A_{40}$ ,  $A_{43}$ ,  $A_{05}$ , and  $A_{33}$  (also the highest by univariate  $|r|$ ) dominate.

### 3 Conclusions

We answered three questions and one bonus question:

1. Several AlphaEarth dimensions carry strong univariate signal for SOC (Method 1;  $|r|$  up to 0.61).
2. Mean log-SOC differs significantly between Grasslands and Croplands (Method 2; Welch’s  $t$ ,  $p < 10^{-9}$ ).
3. The mean-difference and correlation effects are robust to distributional assumptions (Method 3; BCa bootstrap intervals exclude zero and agree closely with the parametric CI).
4. A Random Forest using all 64 dimensions predicts log-SOC with  $R^2 \approx 0.75$  on spatially-held-out blocks — a useful baseline for soil mapping that requires no traditional remote-sensing preprocessing.

#### 3.1 Limitations

Our SOC “ground truth” is itself an interpolated machine-learning product [7], so this work measures alignment between two models rather than against field-measured SOC. The 250 m SOC raster is also coarser than AlphaEarth’s 10 m, and there is a temporal mismatch (OpenLandMap is built on legacy profiles, AlphaEarth is annual 2024). Spatial-block CV mitigates — but does not eliminate — spatial autocorrelation effects.

#### 3.2 Future research

Validating against field measurements (e.g., USDA SSURGO point profiles or ISRIC’s WoSIS database) would convert this from a model-vs-model study into a true accuracy assessment. Extending to multi-depth SOC, pH, and texture, and benchmarking AlphaEarth against a Sentinel-2-only baseline, would isolate the marginal value of the foundation-model embedding.

## References

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